Shop Assistant

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Help & Reference Guide

The Shop Assistant application is a collection of calculations, design, and layout tools for use by the novice woodworker, home owner, and do-it-yourselfer. The various tools available are covered in this guide with photos, sketches, and explanations.

How This App Works

• Most calculators support multiple unit systems (decimal inches, fractional inches, millimeters, feet, etc.) via a unit toggle near the top of the card.

• Many layout calculators (Picket Layout, Spindle Layout, Wall Framing, Diagonal Slats) show a live diagram, and let you select an individual part from a dropdown to see its exact position, length, and cut angles.

• Cards with a “Saved Layouts” section let you save your current inputs under a name and that particular file with its data can be reloaded later.

• Cards with a “Print / Save as PDF” button generate a printable summary of that card's inputs and results, including its diagram where applicable. The PDF can be printed and used at the worksite if desired.

• Use “Data Backup” at the top of the page (Export All Data / Import Data) to move your saved layouts and settings between devices.

01 — Saw Cuts

Setup calculations for common table saw and miter saw operations.

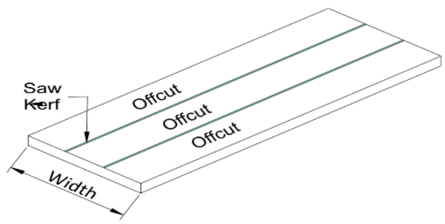
Board Split

Rip a board into equal-width offcuts, accounting for blade kerf.

Usage:

- Select desired units

- Enter Board Width
- Choose number of desired offcuts
- Choose blade
- Blade library can be edited to add/delete blades
- Offcut width and number of cuts are displayed

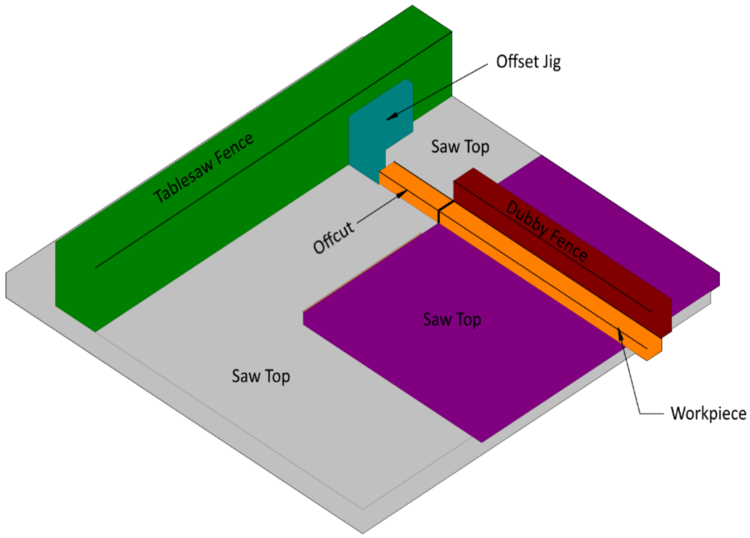


Dubby Sled Offcut

Find the fence setting for cutting to length with Dubby sled

Usage:

- Select desired units
- Enter desired length
- Select the jig
- Jig library can be edited to add/delete blades
- Fence setting is displayed with an image

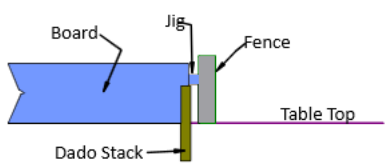


Dado Cut

Plan dado width/depth using multiple passes of a single blade.

Usage:

- Select desired units
- Enter dado stack width
- Enter desired dado width
- Select the jig
- Jig library can be edited to add/delete blades
- Fence setting is displayed

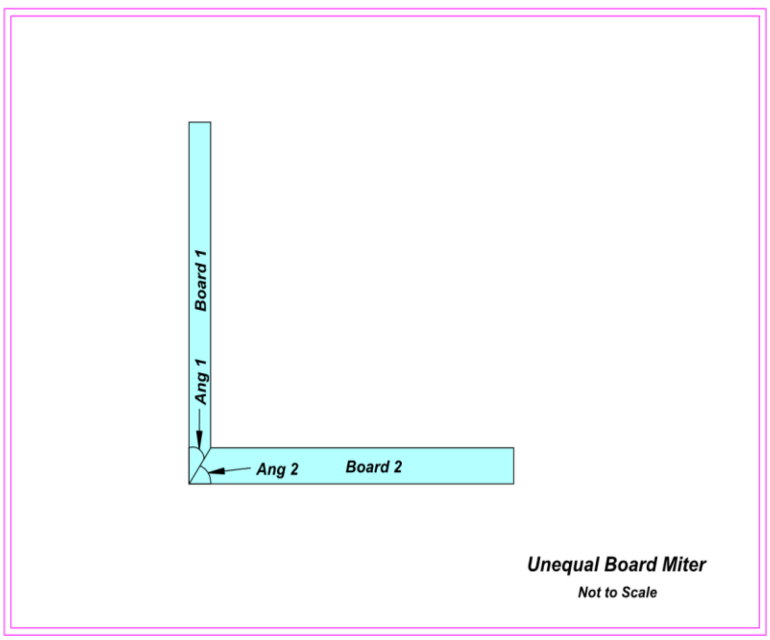


Unequal Board Miter

Miter angles for joining two boards of different widths.

Usage:

- Select desired units
- Enter board A width
- Enter board B width
- Enter corner angle
- Select the jig
- Jig library can be edited to add/delete blades
- Fence setting is displayed

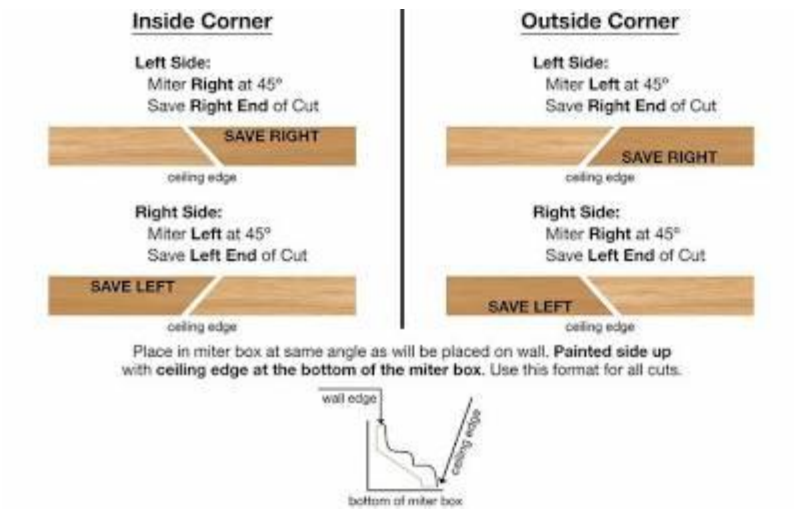


Crown Moulding

Spring angle and blade tilt/miter settings for crown molding. Cuts assume that molding is lying flat on the miter saw table.

Usage:

- Select desired units
- Enter spring angle. If spring angle isn't known, use the spring angle calculator
- Enter corner angle
- Select inside or outside corner
- Miter angle and bevel angle are displayed



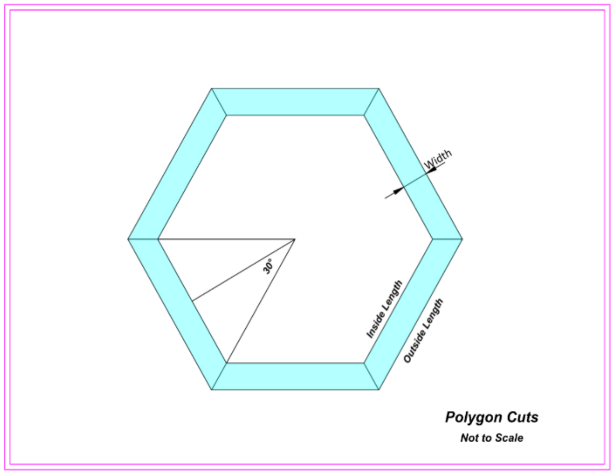
Polygon Cuts

Miter angle, side length, and radius for any regular polygon.

Usage:

- Select desired units
- Enter number of side desired
- Enter board width
- Enter desired side length
- Choose whether side length is inside or outside of the polygon
- Displays the following:
    - Miter angle
    - Interior corner angle
    - Outer side length (long point)

- Inner side length (short point)
- Radius to outer long point



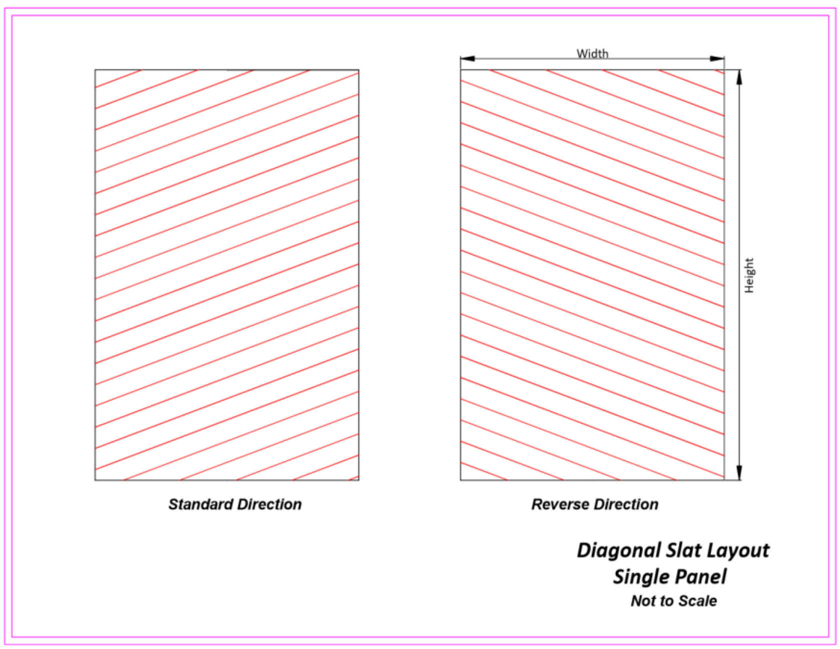
Diagonal Slats

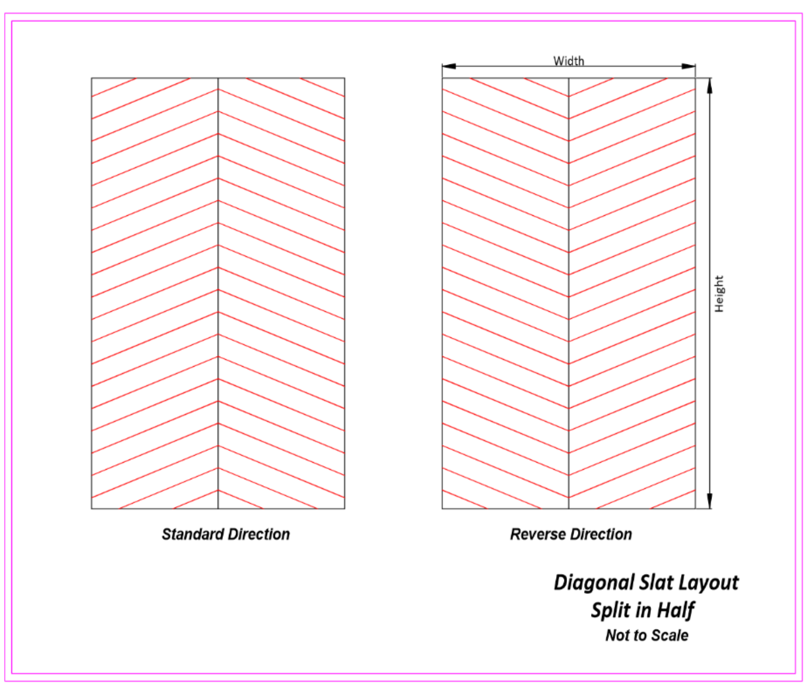
Per-slat blank lengths and miter angles for diagonal slat patterns on a panel, including corner and edge pieces. Slat spacing can be specified.

Usage:

- Select desired units
- Enter panel width
- Enter panel height
- Enter slat angle
- Enter slat width
- Enter gap between slats
- Enter slat direction. Choices are standard and reversed. For standard direction, the slats rise from left to right. For reverse, the slats fall from left to right.
- Enter layout choice of single panel or split in half. If split in half is selected, the panel width is divided and the right half mirrors the left half.
- Displays the following:
  - Number of slats

- Longest blank length
- Total length of slats
- Combo box that displays individual slats by number. When a slat is selected, its length is displayed. In addition, the miter angles for both the left and right ends of the slat are displayed along with the attachment (bottom, top, sides, corner) and miter angle for each end. When a slat is selected, its location is highlighted in the image.





02 — Tool Adjustment

Calibration and fine-tuning calculators for shop machinery and clamp or shim placement

Tablesaw Fence

Fence calibration based on multiple measurements.

Usage:

- Select desired units
- Enter tablesaw fence setting
- Offcut a strip at the fence setting
- Enter 3 caliper readings at points on the offcut
- Displays average caliper reading and difference between fence setting and caliper reading.
- Adjust by setting fence to the displayed result and Zero the fence

5-Cut Adjustment

Classic 5-cut method for squaring a miter/crosscut sled.

Usage:

- Select desired units
- Using a rectangular piece of scrap, mark the sides A, B, C, and D
- Place side A against the fence and cut a thin strip. Then rotate the scrap 90° clockwise, placing the edge just cut against the fence. Continue until all 4 edges have been cut. Then rotate once more and cut a strip about 1in wide. Mark the near end A and the far end B.
- Use a caliper to measure the width of ends A and B and enter that data
- Measure along the fence from the pivot point to the adjustment point and enter that distance in Pivot Length
- Select whether the pivot point is on the right or left of the saw blade.
- The displayed result is the adjustment
- For a quick visual overview of applying this exact formula to a sled fence, check out William Ng's 5-Cut Method.

Planer Adjustment

Planer depth-of-cut and material removal tracking.

Usage:

- Select desired units
- Enter target thickness
- Run a board thru the planer and record caliper readings at 3 points along the board
- Enter these readings
- Display shows
    - Average caliper reading
    - Difference between target thickness and caliper average
    - Approximate number of turns to reach target thickness
    - If the difference is negative, the board is too thin

Router Shims

Shim combinations to fine-tune a router bit and fence to an exact offset.

Usage:

- Select desired units
- Enter the desired distance from the fence to the center of the bit
- Enter the bit size
- The shim distance is displayed in decimal inches, fractional inches, and millimeters
- The closest combination of shims to the shim distance is displayed as well as the total of the shims and the difference between the combination and the shim distance.
- Shim size library can be edited to add and delete shims

Clamps & Shims

Reference calculator for clamping pressure and shim stacking.

Usage:

- Select clamp or shim placement
- Select desired units
- If shims is selected
    - Enter total length
    - Enter end gap
    - Enter shim thickness
    - Displays number of shims and their distance from the pivot point
- If clamps is selected

        - Enter panel length
        - Enter panel width
        - Enter number of boards
        - Displays clamp locations from left end of panel

03 — Design

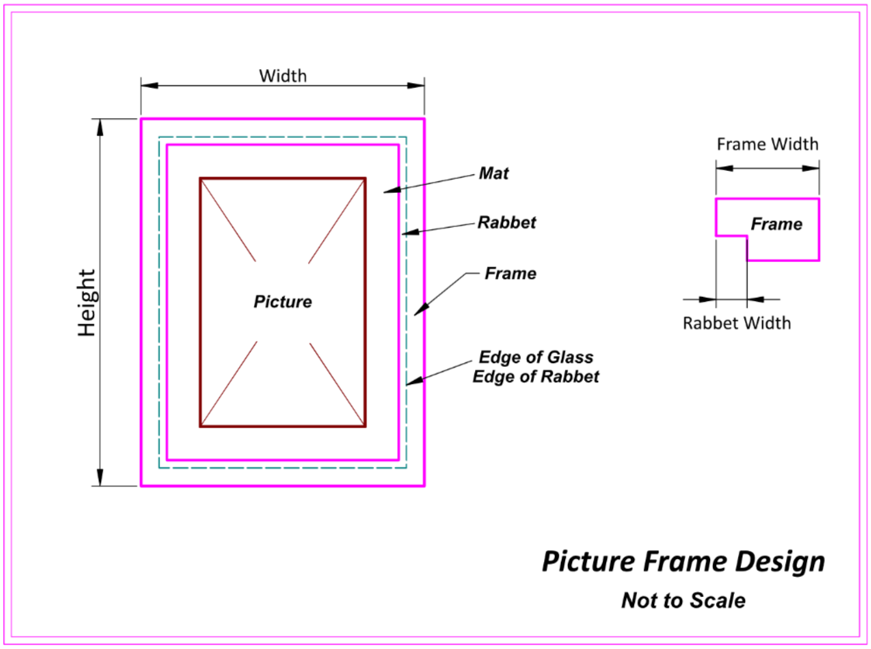
Layout and proportion calculators for various items.

Picture Framing

Frame design and layout, mat design with layout information.

Usage:

- Select desired units
- Enter picture width and height
- Enter glass width and height
- Enter frame and rabbet widths
- Enter picture lap and mat lap. Picture lap is the distance the picture overlaps the mat. Mat lap is the distance mats overlap each other if multiple mats are used.
- Displays the outside width and height of the frame for cutting
- Displays the glass and mat size
- Displays a combo box with choices of 1, 2, or 3 mats. Mat 1 is closest to the glass. As mats are selected the required borders for the picture window are displayed. By default mat borders have an even split for top/bottom and side/side. If a border is edited, the corresponding border is adjusted to maintain the overall mat dimensions.
- Files may be saved for future use



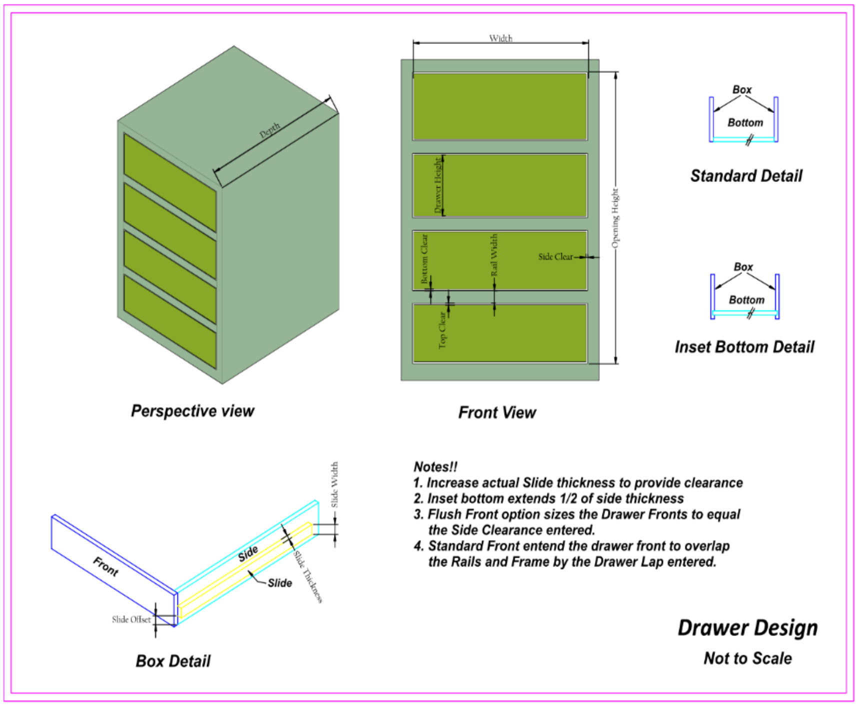
Drawer Design

Drawer design to fit a given space. Layout measurements for rail placement, slide placement, box dimensions accounting for slide clearance and material thickness. Both inset and overlap front are accommodated as well as inset drawer bottoms. Up to 9 drawers can be designed for a given space.

Usage:

- Select desired units
- Enter space height, width, and depth
- Enter top and bottom gap space for drawer clearance
- Enter rail and slide widths
- Enter the slide offset from the bottom of the drawer
- Enter slide thickness
- Enter drawer side and bottom thickness
- If drawers are to be flush mount (inset), check the flush mount box
- Id drawer bottom is to be inset, check the inset bottom box.
- Enter the drawer box space heights. Up to nine drawers can be entered.
- As the drawer heights are entered, the number of drawers entered and the remaining space are displayed. When the space is filled, an alert is displayed that the space is full. No computations are made until the space is full.

- After the space is filled, the user can select rail locations, slide placement, and drawer front heights from the individual combo boxes.
- Also displayed are the dimensions for the box pieces.
- Files may be saved for future use



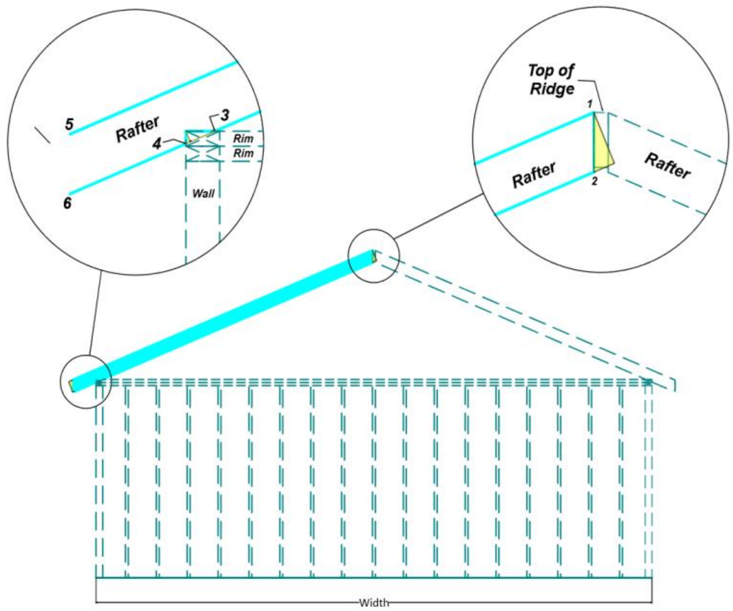
Rafter Design

Rafter design for both A-Roof and Slope Roofs.

Usage:

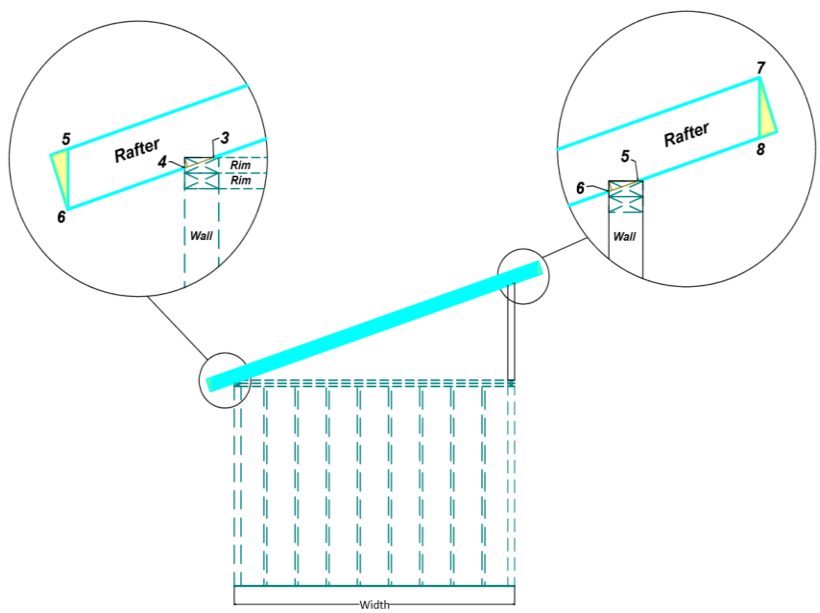
- Select desired units
- Select A-Roof or Slope Roof
- Enter building length and width
- Enter degree rise in decimal degrees
- For A-Roofs, enter soffit width and ridge thickness

- For slope roofs, enter left and right soffit widths
- Select rafter size and on-center spacing
- For slope roofs, enter wall thickness
- Enter notch depth for birdsmouth
- Enter fascia width
- If the notch depth exceeds code, an alert appears. Reduce the notch depth unti the alert disappears.
- Display includes:
    - Minimum stock length
    - Number of rafters
    - Plumb cut angle
    - Level cut angle
    - Seat cut, level component length
    - Rafter layout points
    - Use a square to lay out the birdsmouth cut. Place the with one leg at the notch depth on point 4 and the other leg at the seat cut on point 3. Trace the edges of the square to produce the birdsmouth. If roof is Slope Roof, repeat for points 6 and 5.
    - If the fascia doesn't cover the rafter cut 5-6, do a level cut to suit at point 6
    - Files may be save for future use



A-Roof Section & Details

Not to Scale



Slope Roof Section & Details

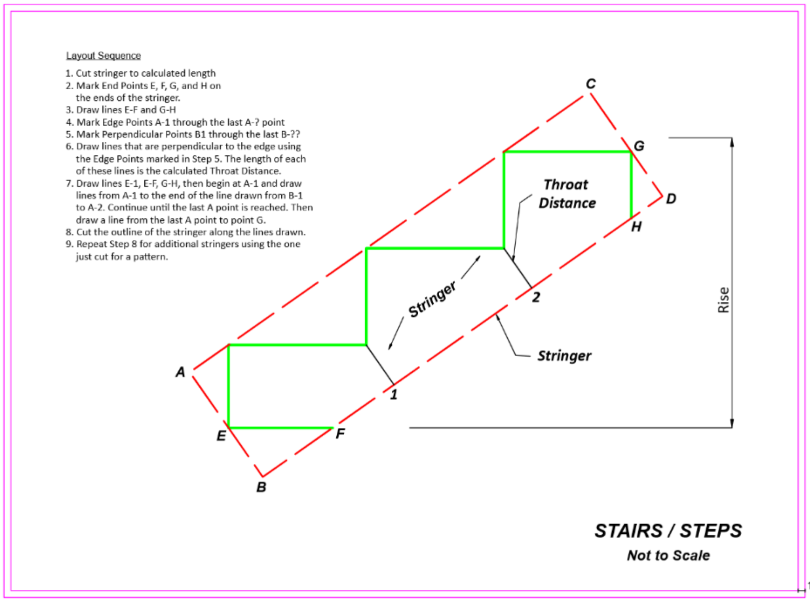
Not to Scale

Stairs & Steps

Designs stairs or steps and provides information for stringer layout. Checks throat distance for code compliance

Usage:

- Select desired units
- Sect tread thickness and width
- Enter total rise and desired nosing
- Enter stringer stock width
- Enter desired riser thickness. If no riser is desired, enter 0
- After above data is entered, a combo box is populated based of a riser height of 7 to 7.75 inches. An additional 3 steps are available should you want less riser height.
- If minimum throat distance is less than 3.5 inches, an alert is displayed. Increase stringer width until alert disappears.
- Based on the number of steps selected, the display shows:
    - *Riser Height*
    - *Stringer Board Length*
    - *Stair Incline Angle*
    - *Throat Distance*
- *Reference points are computed for the stringer edges, top edge points, and bottom edge points*
- *Mark edge points A, B, C, D, F, and G*
- *Mark all A points along top edge of stringer*
- *Mark all B points along bottom edge of stringer.*
- *Draw lines perpendicular to the bottom edge at the B points*
- *Draw lines E-A1, E-F, last A point to G, G-H*
- Begin at A1 and draw lines A1 – B1- A2. Repeat until last A point is reached
- Draw line from last A point to G to complete layout
- Cut along lines to create layout
- Files may be save future use



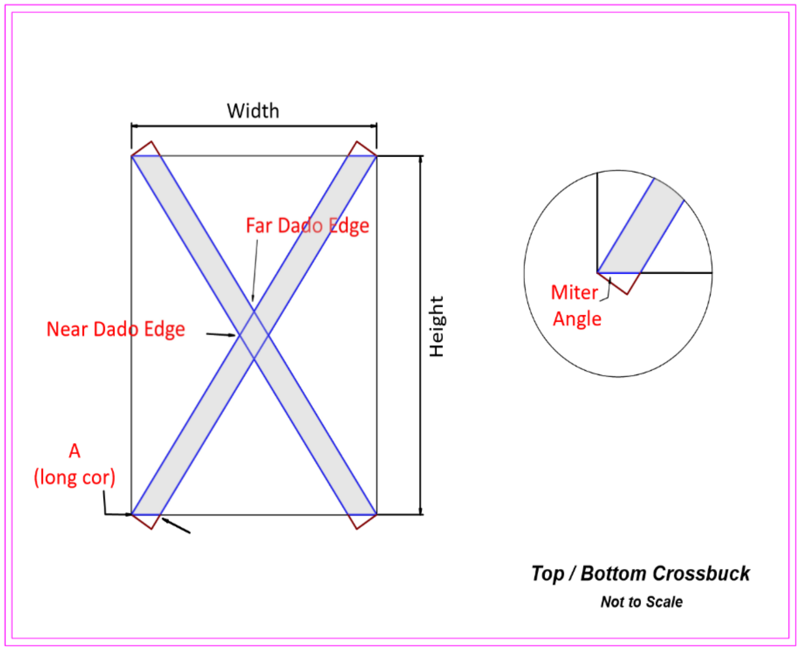
Crossbuck Design

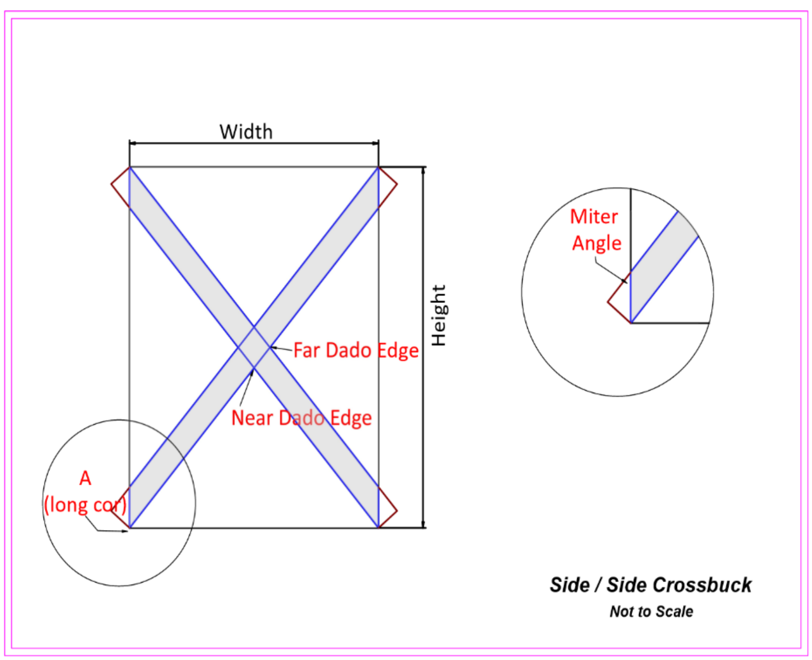
Layout for X-brace (crossbuck) door and gate patterns. Options are for the members to attach the to top/bottom, side/side, or centered on the space edges.

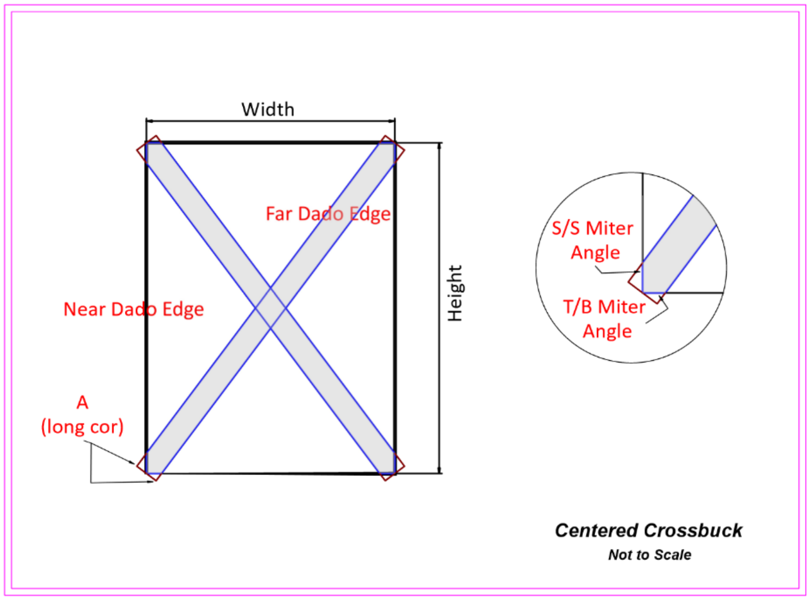
Usage:

- Select desired scale
- Enter space height and width
- Enter member width
- Select type
- Displays:
    - *Member Blank Length*
    - *Miter Angle:*
    - *Long point near and far sistances*
    - *If type is center, there are two miter cuts on each end and there is no long point for marking. In this case, the either edge of the member may be used with distances being measured from the member corner.*
    - *File may be saved for future use*

- *If a dado isn't used, then cut both boards to length. Install the first board, then mark the 4 points on the second board. Connect the near point for each long point to the far point for the opposite long point. Cut the board and install both pieces.*







04 — Layouts

Spacing and positioning calculators for repeating objects.

Object Spacing

Even spacing of objects across a given length. User can specify how many rows, spacing, etc

Usage:

- Select desired units
- Enter space length and object width
- Select type of layout. Choices are objects, spacing, variable space, variable object
- If objects is chosen, enter left gap, right gap, and number of objects.
- Displayed will be the calculated spacing.
- A combo box will populate with a list of object numbers

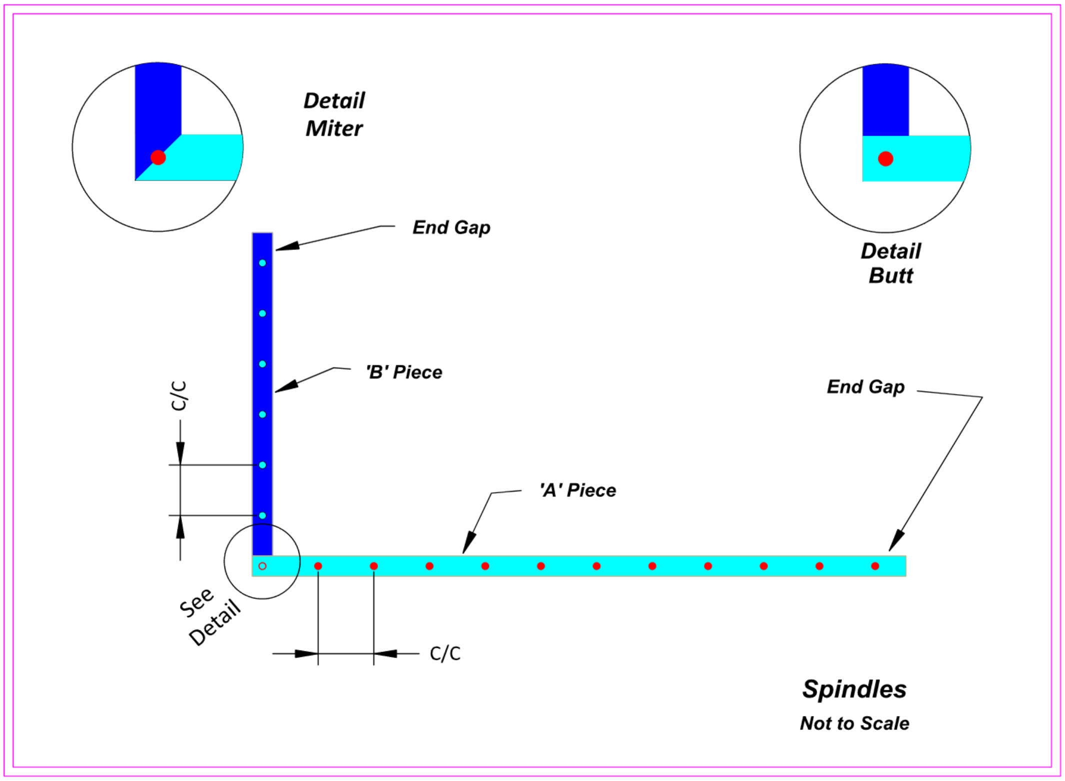
- As objects are selected, their distance from the left end to the left and right edge of the object are displayed. The object's position is highlighted in the image.
- If spacing is selected, the desired spacing is entered and the number of objects is computed. Since this computation will have a remainder, the remainder is didived equally and assigned as the end gaps.
- If variable spacing is selected, an approximate spacing is entered and a trial number of objects is determined, assuming that there will be an object at each end. After the number of objects is determined, the spacing is adjusted to completely fill the space.
- If variable objects is selected, the spacing is held constant and the width of the object changes.
- Files containing the data can be saved, loaded, and deleted
- If lines are desired just enter an object width of 0. Please note that lines don't highlight in the image.

Spindle Layout

Baluster/spindle spacing between two rails, including corner spindles.

Usage:

- Select desired units
- Enter rail A length and rail B length
- Enter gaps at end of rail A and rail B
- Enter rail width and spindle width
- Enter a target spacing and whether the rails will be joined with a miter or butt joint
- After all input is entered, displayed will be :
    - *Total Number of Rail A Spindles:*
    - *Actual Center/Center Spacing — Rail A:*
    - *Total Number of Rail B Spindles:*
    - *Actual Center/Center Spacing — Rail B:*
- *An image is displayed above a 2 combos – one for the A spindles and one for the B spindles*
- *As spindles are selected from either combo, their location is highlighted in the image and their center distance from the rail end is displayed. Distances for the A rail are from the left end of rail A and distances for rail B are measured from the right end of rail B*
- *The layout data can be printed or saved as a PDF for reference in the field.*
- *Files can be saved, loaded, and deleted*



Picket Layout

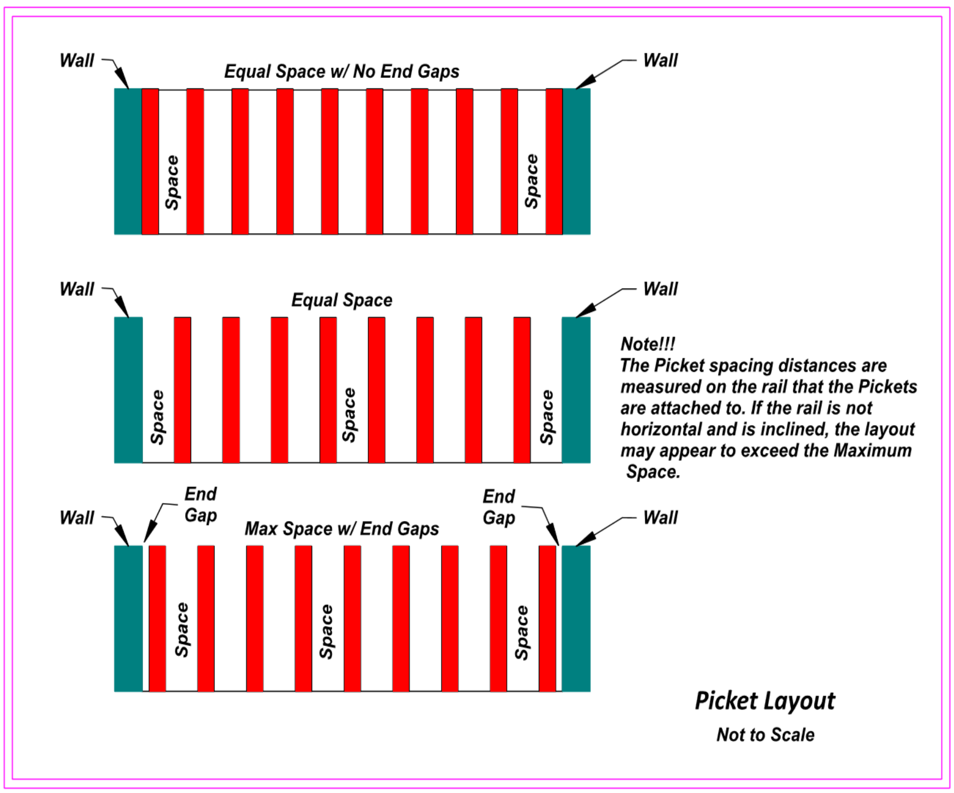
Picket spacing for rectangular and sunburst fence sections and gate patterns.

Usage:

- Select rectangular or sunburst pattern
- Select desired units
- Enter opening width and picket height
- Enter rail incline and maximum spacing. To comply with building codes, the maximum space between pickets is 4 inches.
- Select from equal space with no end gaps, equal space, and maximum space with end gaps for the type.
- When all data is entered, the following is displayed:
  - *Number of Pickets:*
  - *End Gap:*

- *Picket Center/Center:*
- *Space Between Pickets:*

- *An image is also displayed showing the picket layout as well as a combo with the info for each picket. When a picket is selected in the combo, its distance from the left edge of the space to the left edge of the picket is displayed. In addition, the picket location is highlighted in the image.*
- *The layout data can be printed or saved as a PDF for reference in the field.*
- *Files can be saved, loaded, and deleted*



Square-Up (3-4-5 Method)

Verify a corner is square using the 3-4-5 triangle method, with tape-measure running totals.

Usage:

- Select desired units\
- Enter leg A and leg B
- Enter the tape reading at the end of leg A

- Displayed will be :
    - Required diagonal
    - Tape start
    - Reading at corner (end of Leg A)
    - Reading at end of Leg B
    - Reading after diagonal
    - To perform a layout, its best to have 3 people. The person at the end of leg A will hold the tape start and the reading after diagonal on the end of leg A. Another person will hold the tape at distance tape at end of leg A and start of leg B. The third person will hold the tape reading at end of leg B. thus the points are located and they form a 90 degree angle between legs A and B

Wall Framing

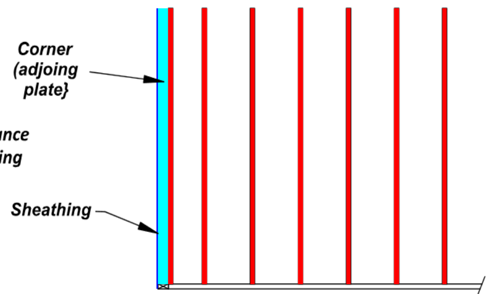
Stud layout for straight and raked (gable-end) walls, including corner framing.

Usage:

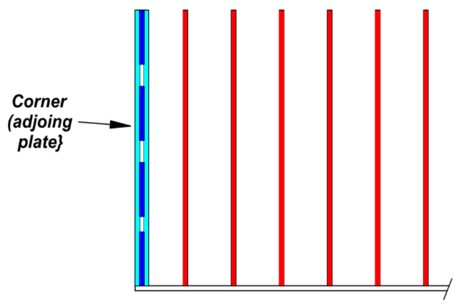
- Select desired units
- Select from either stud walls or rake walls layout.
- Enter foundation length
- Select stud size and on-center measurement
- Select sheathing thickness
- Select whether the wall corners will be placed on the wall ends
- Corners are assumed to require 4 studs per corner
- Ater data has been entered, display values are :
    - *Number of Studs:* ***20***
- An image is also displayed followed by a combo with individual stud info.
- When a stud is selected in this combo, its location is highlighted in the image and its location, measured from the left foundation wall to the left side of the stud.
- In addition to the information entered for stud walls, rake walls requires that the left and right wall heights be entered as well as the level seat length of the rafter birdsmouth that will be installed.
- *The layout data can be printed or saved as a PDF for reference in the field.*
- Data can be saved, loaded, and deleted using the saved walls combo

Notes!!!

1. *It is assumed that the plate that includes the Corner will be assembled first. Corner consists of 2 studs with spacers. Sheathing is installed flush with the ends.*
2. *For the plate that doesn't have a corner, allowance is made in the layout calculation for the sheathing on the adjoining wall.*



Corner not on Plate



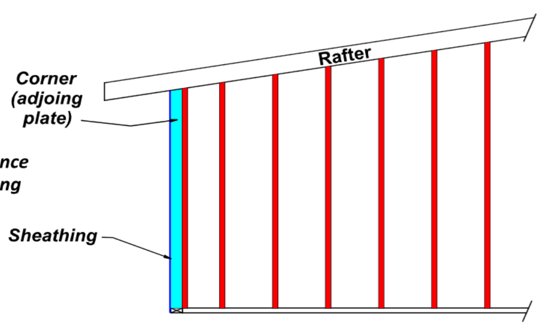
Corner on Plate

Stud Wall Layout

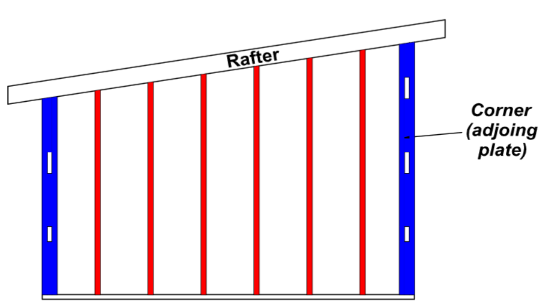
Not to Scale

Notes!!!

1. *It is assumed that the plate that includes the Corner will be assembled first. Corner consists of 2 studs with spacers. Sheathing is installed flush with the ends.*
2. *For the plate that doesn't have a corner, allowance is made in the layout calculation for the sheathing on the adjoining wall.*



Corner not on Plate



Corner on Plate

Rake Wall Layout

Not to Scale

Wheel Segments

Segment angles and lengths for a segmented wheel/circle glue-up.

Usage:

- Select desired units
- Enter segment width and desired outer radius
- Select layout type choice of either segments or inner chord length
- Segments type is used for wheel layout. When the number of segments is entered, a circle is divided into that number of parts. Useful for large radius values
- Inner chord length is useful for laying out a single segment with a large radius.
- When the desired number of layout marks is entered, the following values are displayed
    - Blank length
    - Inner chord length
    - Blank width needed
    - Blank miter angle
- A combo is also populated with layout point numbers. Layout is done from right to left. It is good practice to select an odd number of layout points so that one point will be located at the chord's center
- When a mark number is selected from the combo, the chord distance, inner perpendicular, and outer perpendicular are displayed. Locate all marks prior to cutting the end miters
- After points are marked, use a right triangle to draw perpendicular lines from the marks then mark points on those lines for the inner and outer perpendicular distances.
- Finally use a flexible ruler or other object to draw the arcs

Circular Pattern

Even angular spacing of items around a circle. Rows can be staggered

Usage:

- Select desired units
- Enter outside radius and row spacing
- Enter number of radial lines desired
- Enter number of rows desired
- Finally, enter the stagger desired per row in degrees.
- Additional instructions on laying out by hand are found on the tool page

Rectangular Pattern

Grid spacing of items across a rectangular area with option to stagger alternate rows

Usage:

- Select desired units
- Enter length and width of space
- Enter number of rows and row spacing
- Enter number of columns and amount of stagger
- Displays :
    - Column spacing
    - Stagger offset
    - Total width
    - Row width total
    - Margin (if Width exceeds rows)
    - Total points
- Image with labels is shown to aid in layout

Interval Layout

General repeating-interval spacing calculator. Up to 10 intervals per set and 10 sets may be entered

Usage:

- Select desired units
- Enter tape start
- Select number of intervals and number of sets desired
- When the number of intervals is selected, inputs are created for each interval.
- An image is displayed with the tape readings for each interval of each set

05 — Quantities

Material takeoff calculators for estimating job quantities.

Footprint Shape

Area and perimeter for user entered shapes.

Usage:

- Select desired units
- The shape is drawn by entering a distance then touching one of the direction buttons. When a direction is touched that line appears on an image. Subsequent entries connect to the last line and continue until the shape becomes closed. At that time the shape becomes shaded and its area and perimeter. Options are available to automatically close the shape, delete the last line, and reset the shape.
- Two combos exist below the shape entry. The first is populated with common building materials. The second is populated with common construction materials. As selections change the exact quantity needed and the recommended purchase are displayed. Depending on the construction material selected additional information may be needed before the calculation can be performed.
- Files can be saved, loaded, and deleted

Wall Shape

Wall area accounting for door/window openings.

Usage:

- Select desired units
- Wall shape functions like footprint shape for entering the shape dimensions.
- Enter the wall height and the sum of any voids that exist due to doors/windows/
- Select the desired material and percentage waste and the exact quantity needed and recommended purchase are displayed.
- For paint, the quantities needed is for a single coat.
- Data can be saved to file, loaded later or deleted

Studs

Stud count for a given wall length and spacing, accounting for corners, doors, and windows.

Usage:

- Enter wall length in feet
- Select on-center spacing
- Enter percentage waste
- Enter number of corners, doors, and windows
- 4 studs are allowed per corner and 3 studs for each door and window

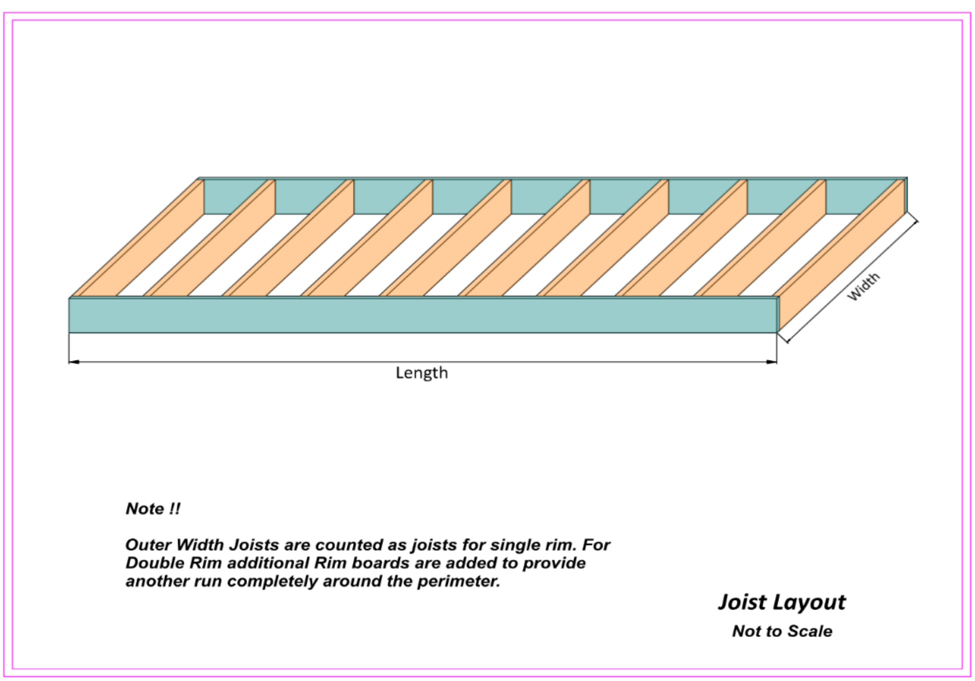
- After calculation, studs needed and number of 12 foot plates are displayed. Plates are assumed to be single on bottom and double on top.

Joists

Joist count and span for floor framing.

Usage:

- Enter foundation length and width
- Select joist on-center and joist length
- Select number of interior supports. To comply with code, the ends of the joists must rest on supports so that maximum span is not exceeded.
- Enter percentage waste
- Displays :
    - Joists needed
    - Rim boards, single rim
    - Rim boards, double rim
- An image is displayed showing the joist layout and interior supports
- The number of interior supports is determined by the maximum span for the joist size and the joist length. Refer to the span tables in the data & information tab for maximum span. Joists must always terminate on a support so if the foundation width is greater than the joist length, a support must be added at the joist length.

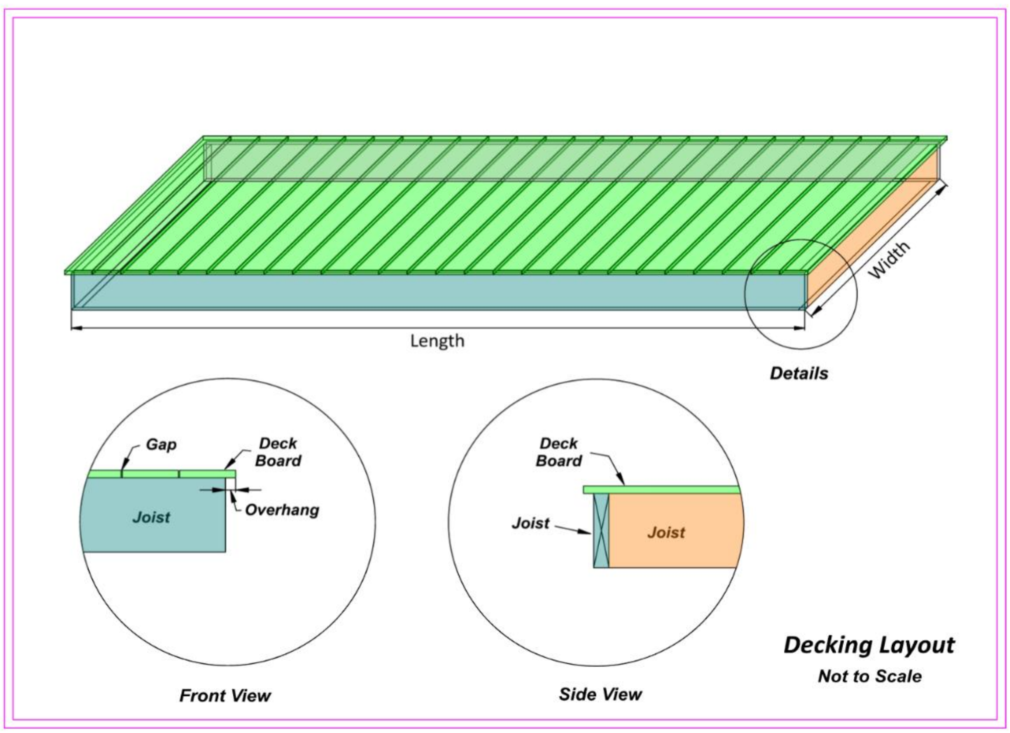


Decking

Decking board quantity and layout for a deck surface.

Usage:

- Enter foundation length and width in feet
- Select board direction
- Enter board width in inches
- Select board length
- Enter gap between boards and overhang
- Select on-center distance and enter percent waste
- Displays :
    - Deck area
    - Board rows
    - Boards needed
- Displays an image of the layout



Roofing

Roofing material quantities based on roof area and pitch.

Usage:

- Enter building length and width
- Select whether roof will be shingles or metal
- Select whether roof is A-roof or sloped roof
- Select roof pitch closest to actual pitch
- Select soffit width for all sides
- Enter percentage waste
- Displays :
    - If roof is shingles
        - Total roof area
        - Plywood sheets required
        - Squares
        - Bundles
        - 15# Felt rolls
        - Roofing tacks (lb)
    - If roof is metal
        - Total roof area
        - Plywood sheets required
        - Number of metal sheets
        - Screws (lb)

Suspended Ceiling

Grid and tile quantities for a drop ceiling.

Usage:

- Select desired units
- Suspended ceiling functions like footprint shape for entering the shape dimensions.
- After room shape has been entered, enter panel length and width
- Select the direction for the panel length to be placed
- Displays :
    - Outline area
    - Number of rows
    - Number of columns
    - Column row gap (each end)
    - Row end gap (each end)

- Number of panels
- Perimeter trim length
- Main beam length
- Cross tee length
- The panel pattern is shown by dashed lines added to the shape image
- File can be saved. Loaded, and deleted

Brick & Block

Brick/block count for a wall of given dimensions.

Usage:

- Select choice of brick or block
- Enter the wall length and height in decimal feet
- If selection is brick choose a brick name from the combo and its length, width, and height are displayed.
- If selection is block, then the combo is populated with common block sizes.
- Selecting a size updates the length, width, and height values
- Enter the mortar joint thickness (normally 3/8in) and percentage waste
- Displayed are :
    - Bricks per course
    - Number of courses
    - Number of bricks
    - Final wall height

Drape Panels

Fabric panel quantities for window treatments.

Usage:

- Units are decimal inches
- Enter opening width and select the distance from 0 to 12 that the panels will be extended past the window opening.
- Select a fullness ratio nd enter the panel width
- Enter a percentage waste
- Display shows “
    - Fabric width needed
    - Panels needed

06 — Calculators

General-purpose reference calculators used across many projects.

Unit Convert

Live conversion between decimal inches, fractional inches, millimeters, feet, meters, gallons/quarts/pints, and pints/ounces/drops.

Usage:

- Enter a value into any input
- Updates the remaining two inputs on that line

Object Clearance

Determines if an object can be rotated inside a space by entering its dimensions.

Usage:

- Enter space width, depth, and height
- Enter object width, depth, and height
- Displays the results of attempting to rotate the object around the X-axis (width), Y-axis (depth), and Z-axis. Displays
    - X-axis clearance
    - Y-axis clearance
    - Z-axis clearance
    - Best clearance distance

Shelf Sag

Predicted shelf deflection under load for a given material and span.

Usage:

- Data is entered in inches and pounds
- Enter shelf length, width. And thickness
- Select material and choose whether ends are fixed or frew
- Enter loading and select whether loading if uniform per foot or point loading
- Displays:
    - Computed sag
    - Acceptable sag

Cabinet Stability

Tip-over risk estimate for a cabinet based on weight, dimensions, and drawer extension

Usage:

- Units are in inches and pounds
- Enter cabinet height and width
- Select number of drawers (1 – 8)
- Next enter drawer data including how far it is extended and the weight in the drawer.
- When all data is available, the stability and factor of safety are calculated

Golden Ratio

Golden-ratio proportion calculator for aesthetic layout.

Usage:

- Select desired units
- Select shape to be analyzed – cube or plane
- Enter the desired measurement then select its cube element – length, width, or height
- If cube was selected the display is :
    - Length
    - Width
    - Height
- If plane was selected the display is
    - Length
    - width

Date Math

Date arithmetic (adding/subtracting days, finding intervals.

Usage:

- if compute ending date is selected:
    - start date may be entered directly or by using the adjacent calendar
    - enter number of desired days _ both + and – days are valid
    - displays ending date
- if compute time difference is selected
    - enter first date directly or with calendar
    - enter second date directly or with calendar

- displays difference in days for the dates
- displays the day difference as years, months, and days

Complementary Angle

Complementary and supplementary angle lookup.

Usage:

- selected desired angle units
- entered angle and its 90 degree complement are displayed in decimal degrees, radians, and degree-minute-second format

Slopes

General slope/angle reference calculator.

Usage:

- enter rise and run to display slope incline
- displays slope incline and slope ratio
- if slope incline is edited, the vertical rise per 100 feet of run is displayed.
- Slope ratio is also updated and displayed as rise per 12 foot run

Wrench Selection

Wrench size lookup for common fastener sizes.

Usage:

- When a selection is made in the S.A.E. combo is made, the equivalent or next larger metric wrench size becomes the selection in the metric combo.
- If a selection is made in the metric combo, the process works in reverse

Slope / Grade

Convert between slope ratio, degrees, and percent grade.

Usage:

- Enter rise in inches and run in feet.
- Displays slope in percent and angle in decimal degrees

Stair Rise & Run

Quick rise/run calculation for stairs.

Usage:

- Enter a value in the total rise input
- Target riser height is calculated and displayed
- Also displayed is number of risers, actual riser height, and suggested tread depth
- If the target riser height is edited, the values are also updated

Survey Levels

Differential calculations for a surveying level, in decimal feet or feet-inch with the ability to add an extra point so the a slope can be calculated

Usage:

- Enter known benchmark elevation and back rod reading
- If the front rod reading is entered or edited, the value displayed in point elevation is updated and displayed
- If the point elevation is entered or edited, the value displayed in the front rod is updated
- An additional point can be calculated by entering a distance and percent grade. The elevation of the additional point is calculated from the first point elevation established. This can be used to check grade on a ditch, etc.
- Once all data is entered, an image is generated showing the layout.

Wire Sizing

Wire gauge selection based on amperage, distance, and voltage drop.

Usage:

- Select wire material and voltage
- Enter load in amps and one way distance for the run
- Based on values entered and selected, the calculations are made and the recommended wire size together with one size smaller and one size larger are added to the compare wire size combo.
- As selections are made in the combo, displayed are :
    - Compliance
    - Maximum amps (ampacity)
    - Calculated voltage drop
    - Percent voltage drop
    - Data can be saved and reloaded later

Curve Solver

General curve/arc geometry solver.

Usage:

- Select desired units
- The 4 curve elements that are calculated are angle, radius, chord, and height. These values are contained in both variable combos.
- When a selection is made in one combo, that selected value is removed from the other combo.
- Values are entered for each selected element and computation is made
- Displayed are :
  - Angle
  - Chrod
  - Height
  - Radius

Ratio Solver

Solves proportions interactively

Usage:

- After 3 values have been entered, the forth value is calculated

07 — Data & Information

Reference tables for common hardware and building specifications.

Wood Screws

Wood screw size and pilot-hole reference.

Usage:

- Select desired scale
- Select screw size
- Displayed results:
  - Clearance hole diameter
  - Hardwood drill size
  - Softwood drill size

        - Head diameter
        - Screw shank diameter

Tap Drill

Tap drill size reference for cutting internal threads.

Usage:

- Select S.A.E. or Metric
- Select screw size
- Select thread type
- displays:
    - Threads per inch
    - Tap drill size
    - Clearance drill size
    - Close drill size

Kreg Screws

Kreg pocket-hole screw selection by material thickness.

Usage:

- If screw length is entered, the recommended material thickness range is displayed
- If material thickness is entered, the recommended screw size is displayed
- If a material thickness is selected that is outside Kreg's recommended material thickness, an alert is displayed

Lag Bolts

Lag bolt sizing and pilot-hole reference.

Usage:

- Select a lag bolt side
- Displays:
    - Hardwood hole size
    - Softwood hole size
    - Countersink size

Fasteners

General fastener reference table.

Usage:

- Select S.A.E. or Metric
- Select fastener size
- Dosplays:
    - Diameter
    - Across flats
    - Across corners
    - Socket clearance
    - Head height
    - Nut height
    - Washer size

Drill Bits

Drill bit size reference and lookup after entering a desired hole size

Usage:

- Select desired units
- Enter hole size
- Displays:
    - Next smaller size
    - Closest size
    - Next larger size

Furniture Dimensions

Standard/ergonomic furniture dimension reference.

Usage:

- Select clamp or shim placement
- Select a category. Categories may be added or deleted
- Select an item and its dimensions are displayed
- Items can be added or deleted

Span Tables

Structural span table reference for joists/rafters based on 2021 IRC code tables

Usage:

- Select rafters or joists
- Select wood type, on-center spacing, and size
- Select live load and dead load
- Displays maximum span

08 — Paint & Chemical Mixes

Mixing and dilution calculators for finishes and chemicals. Includes mixing and a repository for frequently used chemicals

Shellac

Shellac cut ratios and flake mixing

Usage:

- Select flakes or dilute
- For flakes, enter the desired pound cut and volume.
    - Displays the amount of flakes to add and volume
- For dilute, enter the existing pound cut and the volume. Enter the desired cut
    - Displays the volume of alcohol to add

Latex

Latex paint mixing/dilution reference.

Usage:

- Enter desired volume and degree of thinning desired
- Add Floetrol amount to add per ounce
- Displays:
    - Paint needed
    - Water to add

            - Floetrol to add
            - Total mixed volume

Wood Stains

Saved database of wood stain brands, colors and wood species. Brands and stains can be added, deleted, and edited in the manage brands and manage stain names area near the bottom of the page. After some brands and stain names have been entered the appearance of a stain on different wood species.

Usage:

- Select a brand and stain name. then select a wood. If the hex color number and preview change then that particular wood has already been entered. If that is the case, just delete it in the saved entries section and add it again with new values.
- There are several ways to get the hex color number such as using a color picker on a photo or using an app on your phone to get the number.
- When the brand, stain name, wood species, and hex color number are entered, the preview will change color to the hex color number
- Touch add entry to add that brand, color name, wood species, and hex color number are added to the database/

Stain Modification

Adjusting an existing stain mix toward a target result.

Usage:

- Select a brand name, the database is searched and any entered stain names for that brand will populate the stain name combo. When a selection from the stain name combo, the database is searched again and the wood combo is populated with any valid entries
- The original color and diluted color both show the hex color number of the wood species. The diluted color changes as the slider between the boxes is moved.
- Files can be saved, loaded, and deleted

Chemical Mixes

Saved small database of chemical mixes, dilution ratios (e.g. cleaners).

Usage:

- Add your chemicals in the add new chemical section. The chemical name, ounces, and any notes then touch add chemical to add it to the database.
- If a chemical has been selected in the chemical combo, you can select the volume that will be mixed and the strength (1x, 2x, etc). The required volume of chemical is then calculated and displayed

Color Mixing

Suggests up to 3 base colors (from Red, Orange, Yellow, Green, Blue, Indigo, Violet, Purple, White, Black) and mixing percentages/ounces to approximate a target hex color. This is a simplified linear-blend approximation, not a true pigment simulation. The hex color number can be entered or a color picked from the small preview.

Usage:

- Enter a hex color number of use the picker to select a color
- The target preview shows the color selected.
- The closest mix preview displays the closest mix computed.
- Enter the total number of ounces desired
- The calculation shows the colors, their percentage and ounces to produce the closest mix
- Also displayed is the match quality

09 — Settings & Help

app-wide color settings and this help reference.

Colors

Change the app's color theme. Saved automatically and applies everywhere.

Usage:

- Select one of the preset themes or change individual control colors

- Touch reset to revert back to preset themes

Help

Touching the help button will display the help file. It is recommended that you select open help pdf in new tab option. If you are in windows, the new tab appears with the file and its bookmarks. If you are running Android or Ios on a phone, when the tab opens and the file appears without boodmakrs. To display bookmarks, tap the three dots (top right) and select open with and choose a pdf editor that shows bookmarks. Adobe Acrobat Reader works like this: Open the file….click the 3 dots at top right….choose bookmarks & table of contents…..select table of contents… select what you want to view and it will jump to that location.